CAS4133 : Large Language Models

# Final Project

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# Project Overview

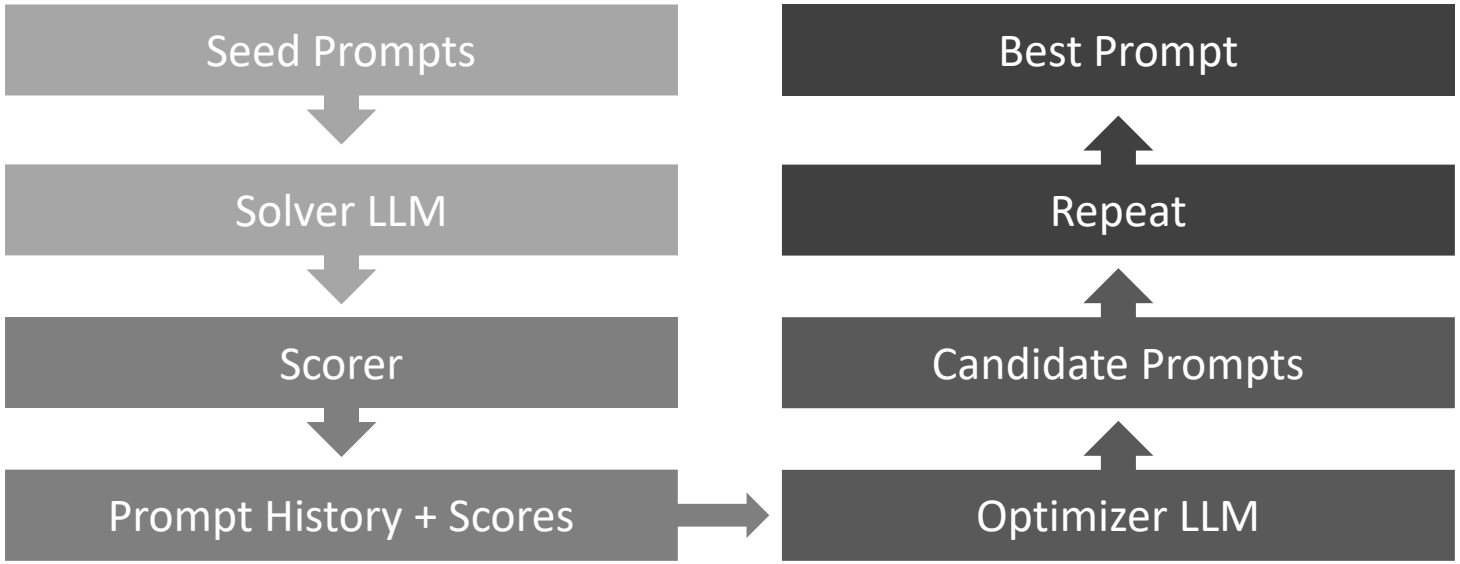
- **Project Name**: Cost- and Format-Aware Prompt Optimization for Large Language Model
- **Motivation**:
  - Prompt engineering is usually manual trial-and-error. However, a good prompt should not only improve accuracy, but also control output format and verbosity.
  - A prompt may solve the problem correctly but fail to follow the required final-answer format, and another prompt may be concise but less accurate. Therefore, prompt quality should be evaluated with multiple objectives.
- **Target Model**:
  - Qwen2.5-3B-Instruct
- **Target Datasets**:
  - GSM8K, SVAMP

# Task Definition

- **Task**: Automatic Prompt Optimization for Math Reasoning
- **Input**:
  - A reasoning dataset
  - Seed prompts
  - A solver LLM
  - An optimizer LLM
- **Output**:
  - An optimized prompt selected by quantitative evaluation
- **Find Prompts that Improve**:
  - Answer accuracy
  - Final-answer format compliance
  - Output efficiency

# Method Overview

- The optimizer LLM receives previous prompts and their scores, then generates new candidate prompts.
- Each candidate is evaluated by the solver LLM on a small validation split.




**Overall Pipeline**

# Scoring Function

- **Vanilla OPRO**:
  - Score = Accuracy
- **Ours**:
  - Score $=$ Accuracy $-\lambda_{\text{format}} \times r_{\text{format}} - \lambda_{\text{token}} \times c_{\text{token}}$
  - $r_{\text{format}}$ denotes format error rate, and $c_{\text{token}}$ denotes normalized token cost
  - We use $\lambda_{\text{format}} = 0.2$, $\lambda_{\text{token}} = 0.1$ and token baseline of 300, and this score favors prompts that are accurate, format-compliant, and concise

# Implementation Details

- The implementation was performed as following:

| Component | Setting |
|---|---|
| Model | Qwen2.5-3B-Instruct |
| Framework | HuggingFace Transformers |
| Datasets | GSM8K, SVAMP |
| Prompt-Search Split | 80 Examples |
| Test Split | 100 Examples |
| Optimization Iterations | 2 |
| Candidates per Iteration | 2 |
| Seed Prompts | Zero-Shot, CoT, Format-Strict |

# GSM8K Validation: Prompt Search

- Optimized Prompt was:
  "**Solve the problem accurately and conclude with the precise line: Final Answer: <number>.**"
- On GSM8K validation, the optimized prompt outperformed all seed prompts in accuracy and final multi-objective score.

| Prompt | Accuracy | Format Error | Avg. Tokens | Score |
|---|---|---|---|---|
| Zero-Shot | 0.5250 | 0.5375 | 87.29 | 0.3884 |
| CoT | 0.4375 | 0.6250 | 99.49 | 0.2793 |
| Format-Strict | 0.5750 | 0.4625 | 83.56 | 0.4546 |
| Optimized | **0.6875** | **0.4000** | **81.94** | **0.5802** |

# GSM8K Test: Accuracy-Only vs. Multi-Objective

- On GSM8K test, multi-objective OPRO achieved higher accuracy and lower format error than accuracy-only OPRO, although its output was slightly longer.

| Method | Selection Objective | Test Accuracy | Format Error | Avg. Tokens |
|---|---|---|---|---|
| Accuracy-Only OPRO | Accuracy | 0.460 | 0.530 | **82.63** |
| Multi-Objective OPRO | Accuracy + Format + Token | **0.520** | **0.490** | 84.18 |

# SVAMP Validation: Prompt Search

- Optimized Prompt was:
  "**Diligently solve the problem and conclude with exactly: Final Answer: <number>.**"
- On SVAMP validation, the optimized prompt improved the multi-objective score while also reducing format error and average output length compared with the seed prompts.

| Prompt | Accuracy | Format Error | Avg. Tokens | Score |
|---|---|---|---|---|
| Zero-Shot | 0.8500 | 0.1250 | 70.73 | 0.8014 |
| CoT | 0.8625 | 0.1125 | 81.38 | 0.8129 |
| Format-Strict | 0.8625 | 0.0750 | 65.19 | 0.8258 |
| Optimized | **0.8750** | **0.0625** | **61.44** | **0.8420** |

# SVAMP Test: Accuracy-Only vs. Multi-Objective

- On SVAMP test, multi-objective OPRO achieved higher accuracy, while accuracy-only OPRO produced slightly lower format error and fewer tokens.

| Method | Selection Objective | Test Accuracy | Format Error | Avg. Tokens |
|---|---|---|---|---|
| Accuracy-Only OPRO | Accuracy | 0.750 | **0.120** | **61.29** |
| Multi-Objective OPRO | Accuracy + Format + Token | **0.800** | 0.140 | 64.61 |

# Overall Results Summary

- Across both datasets, multi-objective OPRO achieved higher test accuracy than accuracy-only OPRO.
- It also improved format compliance on GSM8K, while showing a trade-off on SVAMP.

| Dataset | Method | Test Accuracy | Format Error | Avg. Tokens |
|---|---|---|---|---|
| GSM8K | Accuracy-Only OPRO | 0.460 | 0.530 | **82.63** |
| | Multi-Objective OPRO | **0.520** | **0.490** | 84.18 |
| SVAMP | Accuracy-Only OPRO | 0.750 | **0.120** | **61.29** |
| | Multi-Objective OPRO | **0.800** | 0.140 | 64.61 |

# Demo Video

- Please refer to the attached video.
- *Note*. To prevent the video from being too long, the experiment setting was reduced.

# Limitations and Future Works

- **Limitations**:
  - Prompt search was performed on a small validation split.
  - Results may depend on the sampled examples.
  - The model was relatively small due to limited workspace resources.
  - Multi-objective weights were manually chosen.
- **Future Works**:
  - Tune $\lambda_{\text{format}}$ and $\lambda_{\text{token}}$ automatically.
  - Use larger models or more datasets.
  - Use a separate verifier model for format and answer checking.
  - Apply the framework for non-math tasks.

# Conclusion

- Implemented an OPRO-style prompt optimization application.
- Extended accuracy-only prompt search to multi-objective prompt optimization.
- Evaluated prompts using accuracy, format error rate, and output length.
- Compared multi-objective OPRO with vanilla accuracy-only OPRO.
- Tested the same pipeline on two reasoning datasets: GSM8K and SVAMP.

# Grading Criteria

- (**+10**) Submitted summary slides and codes before deadline
- (**+3**) Implemented OPRO, which was not covered in the lecture
- (**+3**) Included quantitative evaluation results
- (**+2**) Submitted a demo video
- (**+2**) Used two datasets: GSM8K and SVAMP